

Entailment as Plural Modal Anaphora

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I. Introduction.

Meaning vs. Content.

The Partee marble examples:

- (1¹) and (2¹): different meanings (different anaphora licensing potential), but the same content.

(1) ¹ I dropped ten marbles and found all of them, except for one.

² It is probably under the sofa.

(2) ¹ I dropped ten marbles and found only nine of them.

² It is probably under the sofa.

I. Introduction.

Meaning vs. Content.

Entailment is a relation between contents.

(1¹) entails and is entailed by (2¹),
i.e. they express the same content, the same *factual information* (the same information about the world – they have the same truth-conditions).

Meaning is more than factual information: it is also *discourse information* (e.g. anaphoric relations).

I. Introduction.

The Structure of the Presentation.

- I. **Introduction.**
 - II. **The Phenomenon:**
Entailment in Natural Language.
 - III. **The Basic Proposal:**
Entailment as Modal (i.e. Content) Anaphora.
 - IV. **A Dynamic System for Modal Anaphora.**
Defining Truth and Entailment.
- Appendix:** Comparison with Static and Dynamic Definitions of Entailment.

I. Introduction. Meaning vs. Content

Another example: Active vs. Passive Sentences;

- (3¹) and (4¹): different meaning (different anaphora resolution preferences), same content.

(3) ¹ At 1 AM last night, Veronica was yelling at Josephine. ² She was drunk again (i.e. Veronica).

(4) ¹ At 1 AM last night, Josephine was being yelled at by Veronica. ² She was drunk again (i.e. Josephine).

I. Introduction.

Meaning, Content and Context.

Meaning determines content *given a context*.

(5), when uttered on a Thursday in a discussion about John, entails ($\models$) (6).

(5) **He** came back
three days ago.

(6) **John** came back
on Monday.

We need the utterance context to determine the contents of (5) and (6) and the entailment relation.

I. Introduction.

Meaning, Content and Context.

Meaning changes the context.

(7) ¹ A man came in. ² He sat down.

(7¹) changes the context relative to which the content of (7²) is determined: the man that came in is the one that sat down.

I. Introduction. Accessing Content and Determining Entailment.

In sum, to determine entailment relations...

- we need to *access contents* within discourse;
- but: determining content is *context sensitive* and *discourse changes the context*.

I. Introduction.

Why contents and entailment?

- Why do we need explicit access to contents?
- Because various natural language phenomena are **content** sensitive.
- Entailment (although a paradigm example) is only one of them...

I. Introduction.

Questions and Entailment.

Answerhood conditions are content sensitive

Entailment ($\models$) determines what counts as an answer:

ϕ is an answer to question $? \psi$ **iff** there is world w such that $\|\phi\|^M \models \|\psi\|^M, w$
(see Groenendijk & Stokhof (1996), Def. 4.19)

I. Introduction: Presupposition and Entailment

Cleft presuppositions are content sensitive

(8) A: ... was the invitation to New York for which I did not apply. **I was just invited.**

(35 lines later)

A: It was he who invited me.

(see Spender (2000): 2, Example 2)

I. Introduction: Presupposition and Entailment

Factive presuppositions are content sensitive

(9) A: **It** [a violent streptococcus] **was lethal to expectant mothers** with small children. (38 lines later)

A: After all, I mean you can't go down and shop if **you know that you're going to knock out an expectant mother.**

(see Spender (2000): 3, Example 5)

I. Introduction:

Entailment and Discourse Relations.

Discourse Relations are content sensitive,
e.g. **Emphasis**.

(10) H: you can stop right there: **take your money**.

J: **TAKE THE MONEY**.

H: absolutely...

(see Walker (1993): 21, (13))

I. Introduction: Discourse Relations and Entailment.

Emphasis again.

(11) It's unfortunate that **it's cloudy in San Francisco this week**, but **CLOUDY IT IS** – so we might as well go listen to the LSA papers.

(see Walker (1993): 27, (26))

(12) **I don't like to go down that way**. It may be shorter, but **I DON'T LIKE IT**.

(see Walker (1993): 199, (105))

I. Introduction.

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Defining Truth and Entailment.
- Appendix.** Comparison with Static and Dynamic Definitions of Entailment.

II. The Phenomenon. Entailment in Context.

Entailment is a relation between contents.

But we determine contents with respect to a context: (5) entails ($\models$) (6) **if** it is uttered on a Thursday in a discussion about John.

(5) **He** came back
three days ago.

(6) **John** came back
on a Monday.

II. The Phenomenon. Entailment and Anaphora.

Moreover, discourse changes context:

premises change the context with respect to which the content of the conclusion is determined.

(16) **A man** came in. $\models$ (17) **He** entered.

II. The Phenomenon. Entailment and Anaphora.

(16) **A man** came in. $\models$ (17) **He** entered.

(16) entails (17) if the anaphor *he* in (17) refers to the same thing (i.e. the same discourse referent) as the antecedent *a man*.

II. The Phenomenon. Entailment and Plural Anaphora.

(18) **Every man** saw **a woman**. (19) **They** noticed **them**.

(18) entails (19) if the plural anaphors ***they*** and ***them*** refer back to all (salient) men and the women they saw...
... **in a structured way.**

II. The Phenomenon. Entailment and Plural Anaphora.

(18) **Every man** saw **a woman**. (19) **They** noticed **them**.

In a structured way, that is:

Entailment does not obtain if man m_1 saw woman n_1 and m_2 saw n_2 ...

but we interpret (19) as asserting that m_1 noticed n_2 and m_2 noticed n_1 .

II. The Phenomenon. Entailment and Plural Anaphora.

(18) **Every man** saw **a woman**. (19) **They** noticed **them**.

The seeing and the noticing have to be **structured** in the same way.

The **pairing** / **distribution** of individuals in the two sentences has to be the same.

II. The Phenomenon. Entailment and Modal Anaphora.

The anaphoric connections between premises (20¹-20²) and conclusion (21) can be *modal*:

(20) ¹ A wolf **might** come in. ² It **would** see John.

(21) It **would** notice him.

(20¹) introduces a possible scenario: a wolf comes in;
(20²) elaborates on this scenario: said wolf sees John;
(21) is entailed only if it is interpreted with respect to the same possible scenario.

II. The Phenomenon. Entailment and Modal Anaphora.

(20) ¹ A wolf **might** come in. ² It **would** see John.

Modal anaphora is **plural**. (20) says that:

- there is an epistemic possibility p in which a wolf u comes in and sees John;
- any epistemic possibility p in which a wolf u comes in is such that the wolf u sees John.

II. The Phenomenon. Entailment and Modal Anaphora.

(20) ¹ A wolf **might** come in. ² It **would** see John.

Modal anaphora is **structured**.

If we consider possibility p_1 in which wolf u_1 enters
and possibility p_2 featuring intruder u_2 ,
then (20²) asserts that p_1 features the same wolf u_1
(and not u_2 !) seeing John (similarly for p_2 and u_2)

II. The Phenomenon. Entailment and Modal Anaphora.

(20) ¹ A wolf **might** come in. ² It **would** see John.

(21) It **would** notice him.

Entailment involves **structured plural** modal anaphora.

The seeing in (20²) and the noticing in (21) have to be **structured** in the same way.

The **pairing** / **distribution** of possibilities and wolves has to be the same for entailment to obtain.

II. The Phenomenon. Summary.

To determine **entailment** (answerhood, presupposition etc.)
between two contents,

i.e. to determine that one content contains at least as much
factual information as the other, we need:

- to explicitly *store* and *access contents* in discourse;
- to account for the fact that:
 - (a) determining content is *context sensitive*
and the context stores *structured pluralities*;
 - (b) the premises *change the context* of the conclusion.

III. The Proposal.

Entailment as Plural Modal Anaphora.

The crucial observation:

Modal anaphora provides a paradigm for accessing contents and determining relations between them.

III. The Proposal.

Entailment as Plural Modal Anaphora.

Entailment relations hold between suitably selected **plural modal discourse referents**,

which are used to store and access contents in a context sensitive way.

III. The Proposal.

Entailment as Plural Modal Anaphora.

Building on previous work on modal and plural individual anaphora (see Stone (1999) and van den Berg (1996)):

I propose a **new dynamic system** which can represent both the **static contents** and the **dynamic meanings** of utterances in discourse.

IV. A Dynamic System for Modal Anaphora. The Structure of The Section.

The Structure of the Section.

- General features of the system (type logic, information states, discourse referents)
- Why plural information states
- Representing contents and meanings: the **max** (maximizing) operator
- Modal subordination as modal anaphora
- Defining **truth** and **entailment**

IV. A Dynamic System for Modal Anaphora. Many-sorted Type Logic.

Following Muskens (1995), the dynamic system is formulated in **many-sorted type logic**.

Basic types (we ignore the temporal and eventuality domains):

- type t : truth-values;
- type e : individuals;
- type s : models variable assignments;
- type w : possible worlds.

IV. A Dynamic System for Modal Anaphora. Info States and Discourse Referents.

- **info states** $I, J, K, \dots$ are **sets of 'variable assignments'**,
i.e. they are of type st .
(just as in van den Berg (1996))
- an **individual discourse referent** (dref) u is of type se
and it stores a **plurality** with respect to an info state I :
(22) $uI := \{x_e : \exists i_s \in I_{st} (x = ui)\}$
(the subscripts on terms indicate their type)

IV. A Dynamic System for Modal Anaphora. Modal Discourse referents.

- a **modal discourse referent** (dref) ***p*** is of type ***sw*** and it stores a **proposition** (a set of worlds) with respect to an info state ***I***:

$$(23) \ pI := \{w_w : \exists i_s \in I_{st} (w = p\hat{i})\}$$

(the subscripts on terms indicate their type)

IV. A Dynamic System for Modal Anaphora. Why Plural Info States?

Why do we model drefs for pluralities and propositions in this way and not via drefs for sets?

- their type would be $s(et)$ for pluralities of individuals and $s(\mathbf{w}t)$ for propositions.

IV. A Dynamic System for Modal Anaphora. Why Plural Info States?

Because we need to capture **structured intersentential plural anaphora**.

Plural info states (type *st*) are able to store and pass on **the internal distributive structure of pluralities**.

IV. A Dynamic System for Modal Anaphora. Why Plural Info States?

(18) Every man saw a woman.

- it asserts that for each and every man, there is a woman that he saw: m_1 saw n_1 , m_2 saw n_2 etc.
- so, the output info state after processing (18) looks something like this...

IV. A Dynamic System for Modal Anaphora. Why Plural Info States?

l	u_1 (men)	u_2 (women)	...
i_1	$m_1 (=u_1 i_1)$	$n_1 (=u_2 i_1)$	...
i_2	$m_2 (=u_1 i_2)$	$n_2 (=u_2 i_2)$	...
i_3	$m_3 (=u_1 i_3)$	$n_3 (=u_2 i_3)$	...
...	...	...	...

each assignment $i_1, i_2, \dots$ stores a particular man and a particular woman that stand in the *see*-relation.

IV. A Dynamic System for Modal Anaphora. Why Plural Info States?

I	u_1 (men)	u_2 (women)	...
i_1	$m_1 (=u_1 i_1)$	$n_1 (=u_2 i_1)$	...
i_2	$m_2 (=u_1 i_2)$	$n_2 (=u_2 i_2)$	...
i_3	$m_3 (=u_1 i_3)$	$n_3 (=u_2 i_3)$	...
...	...	...	...

for each $i \in I$, the man in i saw the woman in i .

IV. A Dynamic System for Modal Anaphora. Why Plural Info States?

(18) **Every man** saw a woman.

(19) **They** noticed **them**.

Since the plural info state I is passed on to (19), we are able to retrieve the *seeing* structure:

for each $i \in I$, the man in i saw, **hence noticed**, the woman in i (and not some other woman).

IV. Intermediate Summary.

So:

- we are working with an intensional logic with an extra type modeling variable assignments;
- Information states (which store and pass on discourse information) are **sets of variable assignments...**
to model pluralities and their discourse dynamics.

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

Sentence **Contents** (propositions): modal drefs

Sentence **Meanings**: Discourse Representation
Structures (DRSs),
i.e. relations between info states of type $st(stt)$

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

[$u, p \mid \text{come_back}_p\{u\}$] - a DRS,
i.e. a relation over info states of type $st(stt)$

General format:

[**new drefs**, e.g. $u, p \mid$ **conditions**, e.g. $\text{come_back}_p\{u\}$]

Interpretation: $\lambda I/J. \llbracket \text{new drefs} \rrbracket J \ \& \ \text{conditions} J$

i.e. the output state **J** differs from **I** at most with respect to the **new drefs** and each **condition** is satisfied by the output state **J**.

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

(6) **John** came back **on Monday**.

Content: $\{w: \text{come_back}_w(\text{john})\}$

i.e. the *maximal* set of worlds w in which John comes back.

come_back is an intensional property of type $\mathbf{w}(et)$, i.e. it relates worlds and individuals (atomic or not).

see, for example, is an intensional relation of type $\mathbf{w}(e(et))$.

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

(6) **John** came back **on Monday**. $\sim > \{w: come_back_w(john)\}$

We introduce a **max** operator over modal drefs to be able to extract and store this proposition.

Meaning of (6): $\mathbf{max}^p ([\mid come_back_p\{John\}])$

Content of (6): encoded by the modal dref p

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

Interpretation: $\lambda I/J. \llbracket \text{new drefs} \rrbracket J \ \& \ \text{conditions} J$

$\llbracket \mid \text{come_back}_p\{\text{John}\} \rrbracket :=$
 $\lambda I/J. \llbracket \] J \ \& \ \text{come_back}_p\{\text{John}\} J$

where:

- $\llbracket \] J$ is just identity, i.e. $I=J$
- $\text{come_back}_p\{\text{John}\} := \lambda I. \forall i \in I (\text{come_back}_{pi}(\text{John}i))$
- $\text{John} := \lambda i. \text{john} (\text{John} - \text{dref of type se}; \text{john} - \text{type e})$
(see Muskens (1996))

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

$I=J$	p (worlds)	<i>John</i>	...
i_1	$w_1 (=pi_1)$	<i>john</i> ($=ui_1$)	...
i_2	$w_2 (=pi_2)$	<i>john</i> ($=ui_2$)	...
...	...	...	...

$\forall i \in I$ (*come_back* _{pi} (*Johni*)), i.e.

come_back _{w_1} (*john*) , *come_back* _{w_2} (*john*) ...

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

max p ([| *come_back* $_p$ {John}])

max p (D) – 'dynamic λ -abstraction over worlds'

- the 'abstracted variable' is the modal dref p
- the 'scope' is the DRS D
- we extract a set of worlds pJ (where J is the output info state) such that: (a) each $w \in pJ$ 'satisfies' D; (b) pJ is the *maximal* such set.

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

$$\mathbf{max}^p(D) := \lambda IJ. \exists H (\llbracket p \rrbracket H \ \& \ DHJ) \\ \& \ \forall K (\exists H (\llbracket p \rrbracket H \ \& \ DHK) \rightarrow pK \subseteq pJ)$$

- 1st conjunct: we introduce p as a new dref (symbolized by $\llbracket p \rrbracket H$) and make sure each world in pJ satisfies D (by DHJ)
- 2nd conjunct: maximality; any other set pK that satisfies D is included in pJ .

IV. A Dynamic System for Modal Anaphora. Representing Contents and Meanings.

max $^p ([\mid come_back_p\{John\}])$:=

$\lambda IJ. I[p]J \ \& \ pJ \subseteq \{w: come_back_w(john)\}$
 $\ \& \ \forall K (\ pK \subseteq \{w: come_back_w(john)\} \rightarrow pK \subseteq pJ)$

$\lambda IJ. I[p]J \ \& \ pJ = \{w: come_back_w(john)\}$

IV. A Dynamic System for Modal Anaphora. Representing Modal Subordination.

All these ingredients are independently needed for representing modal subordination.

(20) ¹ A wolf **might** come in. ² It **would** see John.

- there is an epistemically accessible possible world w in which a wolf x comes in and sees John;
- any epistemically accessible possible world w in which a wolf x comes in is such that the wolf x sees John.

IV. A Dynamic System for Modal Anaphora. Representing Modal Subordination.

(20) ¹ A wolf **might** come in. ² It **would** see John.

- we need **maximality** (we consider *any* epistemically accessible world in which a wolf comes in);
- we need **structured** pluralities: if wolf x_1 enters in world w_1 and x_2 in world w_2 , sentence (20²) requires w_1 to be such that x_1 (and not x_2 !) sees John.

IV. A Dynamic System for Modal Anaphora. Representing Modal Subordination.

(20) ¹ A wolf **might** come in.

$\mathbf{max}^p ([u | \text{wolf}_p\{u\}, \text{come_in}_p\{u\}, p \subseteq p_o])$

- **might** introduces the *maximal* possibility **p** in which some wolf **u** comes in, i.e. **p** collects any world **w** in which there is a wolf that comes in
- **p_o** provides the contextually specified set of *epistemically accessible* possible worlds.

IV. A Dynamic System for Modal Anaphora. Representing Modal Subordination.

J	p (worlds)	u (wolves)	...
i_1	$w_1 (=pi_1)$	$x_1 (=ui_1)$	...
i_2	$w_2 (=pi_2)$	$x_2 (=ui_2)$	...
...	...	...	...

- we establish a structured correspondence between worlds and intruding wolves via the conditions:

$$wolf_p\{u\}J := \forall i \in J (wolf_{pi}(ui))$$

$$come_in_p\{u\}J := \forall i \in J (come_in_{pi}(ui))$$

IV. A Dynamic System for Modal Anaphora. Representing Modal Subordination.

(20) ² It **would** see John.

[| $see_p\{u, John\}$]

- **it** is an individual anaphor, referring back to the intruding wolf **u**; **would** is a modal anaphor, referring back to the maximal possibility **p**
- sentence (20²) simply tests that, in each world *w* in possibility *p*, the corresponding wolf *x* sees John
- Stone (1999) first proposed to analyze modal subordination as anaphora to modal objects (namely accessibility relations)

IV. Intermediate Summary.

- We use **modal drefs** to store propositions.
 - We use **maximal** modal drefs to store contents and maximal possibilities (see modal subordination).
 - We use plural info states to store **structured propositions / contents** and **structured sets of individuals...**
- ...so that we can predict the correct interaction between modal and individual anaphora, i.e. to account for modal subordination and entailment.

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

TRUTH.

A DRS **D** interpreted relative to a dref **p** is *true* with respect to an info state **I** and a world **w** iff

there is an output state **J** s.t. $\max^p(\mathbf{D}) \mathbf{I} \mathbf{J} \ \& \ w \in p\mathbf{J}$.

i.e. $p\mathbf{J}$ is the proposition / content expressed by **D** in context **I** and we check that this content (proposition) is true in world **w**.

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

e.g. take the DRS:

(24) [| *come_back*_p{*u*}]

and the input info state *I* is such that *ul*=*{john}*

(intuitively, *D* could be the representation of the sentence *He came back* in a context in which we are talking about John)

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

(24) is *true with respect to I and a world w*
iff $\exists J (\mathbf{max}^p ([\mid \text{come_back}_p\{u\}]) \mid J \ \& \ w \in pJ)$

iff $\exists J (\mid [p]J \ \& \ pJ = \{w' : \text{come_back}_{w'}(\text{john})\} \ \& \ w \in pJ)$

iff $w \in \{w' : \text{come_back}_{w'}(\text{john})\}$

iff $\text{come_back}_w(\text{john})$

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

ENTAILMENT.

A DRS **D** interpreted relative to p entails ($\models$)
a DRS **D'** also interpreted relative to p
with respect to an info state / **iff**

for any intermediate state **K** s.t. $\max^p(\mathbf{D}) \models \mathbf{K}$,
there is an output state **J**

s.t. $\max^p(\mathbf{D}') \models \mathbf{KJ}$ and $p\mathbf{K} \subseteq p\mathbf{J}$.

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

ENTAILMENT.

$D \models D'$ with respect to an info state I iff
for any intermediate state K s.t. $\max^p(D) IK$,
there is an output state J s.t. $\max^p(D') KJ$ and $pK \subseteq pJ$.

i.e. pK is the content expressed by D with respect to the input context I , pJ is the content expressed by D' with respect to the intermediate context K and the content of D is at least as informative as the content of D' .

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

- we have two information states K and J to store the contents of the *two* DRSs D and D' .
- we have only one modal dref p to encode that both the premises and the conclusion are interpreted relative to the same factual information data base, i.e. they form *one* argument.

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

(16) **A man** came in. $\models$ (17) **He** entered.

(29) $[u \mid \text{man}_p\{u\}, \text{come_in}_p\{u\}] \models$ (30) $[\mid \text{enter}_p\{u\}]$

(16) entails (17) with respect to info state / **iff**

$\forall x_e \forall w (\text{man}_w(x) \ \& \ \text{come_in}_w(x) \rightarrow \text{enter}_w(x))$

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

(16) entails (17) with respect to info state / iff

$$\forall K (\mathbf{max}^p ([u \mid man_p\{u\}, come_in_p\{u\}]) IK \\ \rightarrow \exists J (\mathbf{max}^p ([\mid enter_p\{u\}]) KJ \& pK \subseteq pJ))$$

$$\forall f_{we} (\forall w \in S (man_w(fw) \& come_in_w(fw)) \rightarrow S \subseteq \{w': \\ enter_{w'}(fw') \}), \\ \text{where } S = \{w'': \exists x_e (man_{w''}(x) \& come_in_{w''}(x)) \}$$

$$\forall x_e \forall w (man_w(x) \& come_in_w(x) \rightarrow enter_w(x))$$

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

(16) **A man** came in. $\models$ (17) **He** entered
(29) $[u \mid \text{man}_p\{u\}, \text{come_in}_p\{u\}]$ (30) $[\mid \text{enter}_p\{u\}]$

We are able to capture the entailment between (16) and (17) because we can capture the anaphoric dependency between **he** in (17) and **a man** in (16),

i.e. because our dynamic system captures the dependency between the **dref u** in (29) and the **dref u** in (30).

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

Similarly, we are able to capture the **structured** anaphoric dependencies and the entailment between (18) and (19):

(18) Every man saw a woman.
(19) They noticed them.

(*every* is represented as: **max** u ($[\mid \text{man}_p\{u\}]$))

and between (20²) and (21): ((20¹) A wolf might come in.)

(20²) It would see John. (21) It would notice him.

IV. A Dynamic System for Modal Anaphora. Defining Truth and Entailment.

(20²) It would see John. $\models$ (21) It would notice him.

(31) $[\mid \text{see}_p\{u, \text{John}\}] \models (32) [\mid \text{notice}_p\{u, \text{John}\}]$

(31) entails (32) with respect to info state / **iff**

$\forall K (\mathbf{max}^p ([\mid \text{see}_p\{u, \text{John}\}]) IK$
 $\rightarrow \exists J (\mathbf{max}^p [\mid \text{see}_p\{u, \text{John}\}]) KJ \ \& \ pK \subseteq pJ)$

$\forall x_e \in u / \forall w (\text{see}_w(x, \text{john}) \rightarrow \text{notice}_w(x, \text{john}))$

IV. Summary.

To determine **entailment** (answerhood, presupposition etc.)
between two contents,

i.e. to determine that one content contains at least as much
factual information as the other, we need:

- to explicitly *store* and *access contents* in discourse;
- to account for the fact that:
 - (a) determining content is *context sensitive*
and the context stores *structured pluralities*;
 - (b) the premises *change the context* of the conclusion.

IV. Summary.

In the present system...

- we explicitly *store* and *access contents* in discourse via **maximal plural modal drefs**
- we account for the fact that determining content is *context sensitive* and the context stores *structured pluralities* by using **plural information states**
- we account for the fact that the premises *change the context* of the conclusion by using a **dynamic system** and **interpreting the conclusion in the context provided by the premises**

V. Comparison with Alternative Definitions of Entailment.

- static possible worlds semantics defines entailment as proposition / content inclusion.
- this definition is intuitively appealing, since it distinguishes discourse information and factual information and acknowledges that entailment is a relation between contents.
- however, static systems cannot capture the anaphoric dependencies between premise and conclusion.

V. Comparison with Alternative Definitions of Entailment.

Groenendijk & Stokhof (1991)

A DRS **D** entails ($\models$) a DRS **D'** iff $\forall I \forall K (D/IK \rightarrow \exists J (D'/KJ))$

This definition universally quantifies over the input state I , hence it cannot account for the fact that (5), when uttered on a Thursday in a discussion about John, entails (6):

(5) **He** came back
three days ago.

(6) **John** came back
on Monday.

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Why not remove the quantification over the input state I ?

A DRS $\mathbf{D}$ entails ($\models$) a DRS $\mathbf{D}'$ wrt an input state I **iff**

$$\forall K (D/IK \rightarrow \exists J (D'KJ))$$

If the first definition was too restrictive, this one is too lax.

It predicts (35) entails (36) when interpreted in a context I in which John is both happy and tired.

(35) John is happy. (36) John is tired.

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That is, if $pI \subseteq \{w: \text{happy}_w(\text{john}) \ \& \ \text{tired}_w(\text{john})\}$,

then D entails D' with respect to / because the following formula is true:

$$\forall K ([\mid \text{happy}_p\{\text{John}\}] IK \rightarrow \exists J ([\mid \text{happy}_p\{\text{John}\}] KJ))$$

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Our definition of entailment:

- has the intuitive appeal of the static definition since it distinguishes contents and defines entailment as content inclusion.
- makes the correct predictions with respect to (5)-(6) and (35)-(36) because it is intermediate in 'strength' between the two; the definition is in fact equivalent to...

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A DRS **D** interpreted relative to p entails ($\models$) a DRS **D'** also interpreted relative to p wrt an info state **I** **iff**

for any intermediate state K s.t. $([p]; D) \models K$,
there is an output state J s.t. $D' \models KJ$.

i.e. $\forall I' (\models [p]I' \rightarrow \forall K (D \models I'K \rightarrow \exists J (D' \models KJ))$

we preserve the input context I up to the dref p .

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- Although this definition does not transparently exhibit the connection between truth and entailment...
- This definition shows why **entailment is analyzed as modal anaphora**

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Entailment As Modal Anaphora:

- first, the premise is satisfied by many propositions, all subsets of the **content** of the premise as defined above.
- we non-deterministically store in the modal dref p one such proposition.
- when we interpret the conclusion relative to p , i.e. as **anaphoric** to p , the proposition stored in p is guaranteed to satisfy the conclusion.

A decorative graphic on the left side of the slide. It consists of a light green L-shaped bar that starts from the top left and extends downwards. A dark blue horizontal bar is positioned across the middle of the slide, overlapping the green bar. The text "Thank you!" is written in a dark blue font, positioned to the right of the green bar and below the dark blue bar.

Thank you!